

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 40 | Date: 12-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (5 Questions - 10 Marks)

Q1. [2 marks] (Medium)

Here are 10 unique exam questions for your Data Structures Mid Term Examination:

Q2. [2 marks] (Easy)

Which of the following is a characteristic component of a node in a singly linked list?

Options:

A) A data field and a pointer to the previous node. B) A data field and a pointer to the next node. C) Only a data field. D) Only a pointer to the next node.

Q4. [2 marks] (Easy)

Which principle describes the order in which elements are added to and removed from a stack?

Options:

A) First-In, First-Out (FIFO) B) Last-In, First-Out (LIFO) C) Last-In, Last-Out (LILO) D) First-In, Last-Out (FILO)

Q7. [2 marks] (Easy)

Which of the following is the fundamental principle used for managing elements in a queue data structure?

Options:

A) Last-In, First-Out (LIFO) B) Random access C) First-In, First-Out (FIFO) D) Priority-based access

Q8. [2 marks] (Medium)

What is the average and worst-case time complexity for searching an element in a singly linked list?

Options:

A) $O(1)$ B) $O(\log n)$ C) $O(n)$ D) $O(n \log n)$

Section 2: Short Answer (6 Questions - 30 Marks)

Q3. [5 marks] (Medium)

Explain two key differences between a dynamically allocated singly linked list and a static array in terms of memory allocation and insertion/deletion operations.

Q5. [5 marks] (Medium)

Describe the behavior of the push() and pop() operations in a stack data structure. What happens to the stack's top pointer during these operations?

Q6. [5 marks] (Hard)

Outline the steps required to delete a node from the middle of a singly linked list, given a pointer to the node preceding the one to be deleted. Assume the list is not empty and the node to be deleted exists.

Q9. [5 marks] (Medium)

Explain how the enqueue() operation works in a linear queue implemented using an array, focusing on how the rear pointer is managed.

Q10. [5 marks] (Hard)

Define Stack Underflow and Stack Overflow conditions. How can an implementer prevent or handle these conditions in a fixed-size array-based stack?

Q11. [5 marks] (Hard)

What is the primary advantage of using a Circular Queue over a Linear Queue implemented using an array? Illustrate with a brief example.

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